Implementing Direct Mean Moderation for Biomarker-Treatment Interactions: A Worked Example in Signal Calibration

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Abstract

This document traces four attempts to implement direct mean moderation (Architecture A) for the biomarker-treatment interaction in the `pmsimstats-ng` clinical trial simulation framework. Three attempts produced incorrect behavior due to scaling errors: one generated no signal during off-drug periods, another generated signal that decayed too rapidly, and a third generated signal so large that power reached 1.00 under all conditions. The correct formulation uses standardized biomarker values, a binary on-drug indicator, and scaling by the biological response standard deviation to produce effect sizes that are calibrated against the existing multivariate normal (MVN) architecture. We present the mathematical derivation, explain why each incorrect attempt failed, and provide empirical verification. The narrative is intended to illustrate how dimensional analysis and calibration against a known reference implementation can guide the development of simulation components.

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1 Introduction

1.1 The problem

Consider a clinical trial in which a baseline biomarker B_i (e.g., resting blood pressure) is hypothesized to predict which participants will benefit from a drug. A simulation framework generates synthetic trial data and estimates the statistical power to detect this biomarker-treatment interaction. How should the simulation encode the relationship between biomarker and drug response?

There are two fundamentally different approaches, which we call Architecture A and Architecture B. This document is concerned with the implementation of Architecture A.

1.2 Architecture B: differential correlation (MVN)

The existing implementation in `pmsimstats-ng` generates all participant-level variables: biomarker (B_i), baseline symptom score (BL_i), and response components at each timepoint (BR_{it}, TV_{it}, PB_{it}) from a single multivariate normal (MVN) draw:

$$\mathbf{X}_i \sim \mathrm{MVN}(\boldsymbol{\mu}, \Sigma) \tag{1}$$

The biomarker-treatment interaction is encoded in the covariance matrix Σ. Specifically, the correlation between the biomarker and the biological response (BR) component depends on treatment status:

$$\mathrm{Cor}(B_i, \mathrm{BR}_{it}) = \begin{cases} c_{bm} & \text{if on drug at time } t \\ c_{bm} \cdot e^{-\lambda \cdot t_{sd}} & \text{if off drug (with decay)} \\ 0 & \text{if never treated} \end{cases} \tag{2}$$

where c_{bm} is the biomarker-BR correlation parameter, λ is the correlation decay rate, and t_{sd} is time since drug discontinuation.

The interaction signal lives in the *second moment* (covariance) of the joint distribution. It is not visible in the population means; all participants have the same expected BR at a given timepoint regardless of their biomarker value. The signal emerges only in the conditional distribution: given that participant i has biomarker value b, their expected BR is shifted:

$$E[\mathrm{BR}_{it} \mid B_i = b, \text{ on drug}] = \mu_{\mathrm{BR}}(t) + c_{bm} \cdot \frac{\sigma_{\mathrm{BR}}}{\sigma_B} \cdot (b - \mu_B) \tag{3}$$

This is a standard property of the multivariate normal distribution (see, e.g., Mardia et al., 1979, Chapter 3).

1.3 Architecture A: direct mean moderation

Under Architecture A, the biomarker enters the *mean structure* directly. The BR component for participant i at timepoint t is:

$$\mathrm{BR}_{it} = \mathrm{BR}_{it}^{\mathrm{base}} + \beta_{bm} \cdot B_i^* \cdot D_{it} \cdot \sigma_{\mathrm{BR}} \tag{4}$$

where $\mathrm{BR}_{it}^{\mathrm{base}}$ is drawn from the MVN (with no BM-BR correlation), $B_i^* = (B_i - \mu_B)/\sigma_B$ is the standardized biomarker, D_{it} is a binary drug indicator (1 if on drug, 0 otherwise), and σ_{BR} is the BR standard deviation from the response parameters.

The interaction is *deterministic*: given a participant's biomarker value and treatment status, the BR shift is fully determined. The signal lives in the *first moment* (conditional mean).

1.4 Why the distinction matters

Under drug carryover, the drug effect persists into off-drug periods and gradually decays. Under Architecture B, the correlation between biomarker and BR during off-drug periods also decays (Equation 2), eroding the signal that the analysis model uses to detect the interaction. Under Architecture A, the moderation applies only during on-drug timepoints and is absent during off-drug timepoints regardless of carryover; the signal is binary (present or absent), not graded. This architectural difference produces qualitatively different conclusions about statistical power under carryover (see the companion white paper, `02-dgp-mean-moderation-vs-mvn.md`).

2 The calibration problem

Before describing the implementation attempts, we establish the calibration target. For the two architectures to be comparable, they should produce similar effect sizes for the same value of c_{bm}. Under Architecture B, the conditional expectation (Equation 3) tells us that a one-standard-deviation shift in the biomarker produces a BR shift of:

$$\Delta_{\mathrm{BR}} = c_{bm} \cdot \sigma_{\mathrm{BR}} \tag{5}$$

For the parameters used in the Hendrickson et al. (2020) framework ($c_{bm} = 0.45$, $\sigma_{\mathrm{BR}} = 8$), this is $0.45 \times 8 = 3.6$ units of BR per SD of biomarker. Any correct Architecture A implementation should produce a shift of approximately the same magnitude.

We can verify this empirically. Drawing 500 participants under Architecture B with $c_{bm} = 0.45$, the regression of mean on-drug BR on the biomarker yields a slope corresponding to 3.48 BR units per SD of biomarker, with residual SD of 3.89, consistent with the theoretical prediction.

3 Four implementation attempts

3.1 Attempt 1: multiplicative scaling

The most intuitive approach is to scale each participant's BR by a factor that depends on their biomarker value:

$$\mathrm{BR}_{it}^{\mathrm{mod}} = \mathrm{BR}_{it}^{\mathrm{base}} \cdot (1 + \beta_{bm} \cdot B_i^*) \tag{6}$$

```
dat[, (br_col) := get(br_col) * (1 + beta_bm * bm_centered)]
```

Why it fails. During off-drug timepoints, the base BR value $\mathrm{BR}_{it}^{\mathrm{base}}$ is near zero (the Gompertz response function evaluates to approximately zero when time-on-drug is zero). Multiplying a near-zero value by $(1 + \text{small})$ produces a near-zero result. The moderation signal is proportional to the *magnitude* of the base BR, not to the drug exposure status. Consequently, the analysis model sees strong biomarker-BR association during on-drug periods (where base BR is large) and

essentially no association during off-drug periods, precisely mimicking Architecture B's correlation erosion.

Result. Power dropped approximately 60% under carryover, indistinguishable from Architecture B.

Lesson. Multiplicative moderation conflates the interaction signal with the response magnitude. When the response is near zero (off-drug), the signal vanishes regardless of the moderation strength.

3.2 Attempt 2: additive with drug exposure decay

To address the zero-base problem, the formulation was changed to an additive formulation where the moderation signal tracks the pharmacokinetic drug exposure level:

$$\mathrm{BR}_{it}^{\mathrm{mod}} = \mathrm{BR}_{it}^{\mathrm{base}} + \beta_{bm} \cdot B_i^* \cdot \phi(t) \cdot \sigma_{\mathrm{BR}} \tag{7}$$

where the drug exposure function is:

$$\phi(t) = \begin{cases} 1 & \text{if on drug} \\ \left(\frac{1}{2}\right)^{t_{sd}/t_{1/2}} & \text{if off drug with carryover} \\ 0 & \text{if never treated} \end{cases} \tag{8}$$

```
drug_exposure <- (1/2)^(tsd / t1half)
dat[, (br_col) := get(br_col) +
  beta_bm * bm_centered * drug_exposure * br_sd]
```

Why it fails. The drug exposure function decays exponentially during off-drug periods. For the crossover design with $t_{1/2} = 1.0$ week, the drug exposure at the four off-drug timepoints (each 2.5 weeks apart) is 0.18, 0.031, 0.006, and 0.001. The additive moderation signal at these timepoints is $0.45 \times 1 \times 0.18 \times 8 = 0.65$ BR units at the first off-drug timepoint, falling to 0.004 by the fourth. Against per-timepoint noise of $\sigma_{\mathrm{BR}} = 8$, this signal is undetectable.

Result. Power under carryover remained low, comparable to Architecture B.

Lesson. Tracking drug exposure decay in the moderation signal reintroduces the very phenomenon that Architecture A is intended to avoid. If the moderation decays with the drug, the architecture is functionally equivalent to Architecture B from the analysis model's perspective.

3.3 Attempt 3: additive with time-on-drug

We then implemented the formulation found in the original 2025 tidyverse codebase (`pmsimstats2025/analysis/scr` initial commit `957c8bd`):

$$\mathrm{BR}_{it}^{\mathrm{mod}} = \mathrm{BR}_{it}^{\mathrm{base}} + B_i \cdot \mathrm{tod}_t \cdot \beta_{bm} \tag{9}$$

where tod_t is the cumulative time on drug at timepoint t (in weeks), and B_i is the *raw* (uncentered, unstandardized) biomarker value.

```
dat[, (br_col) := get(br_col) + bm * tod * beta_bm]
```

Why it fails. The biomarker in the Hendrickson framework has mean $\mu_B \approx 124$ and SD $\sigma_B \approx 15$ (systolic blood pressure). The cumulative time-on-drug reaches 10–20 weeks. The moderation shift for a typical participant at the last on-drug timepoint is:

$$124 \times 10 \times 0.45 = 558 \text{ BR units} \tag{10}$$

The BR noise SD is 8. The signal-to-noise ratio is approximately 70:1. Statistical power is 1.00 under all conditions, not because Architecture A is resilient to carryover, but because the signal is so absurdly large that no amount of noise or carryover could obscure it.

Result. Power = 1.00 everywhere. This was the behavior observed in the original 2025 codebase, which was incorrectly interpreted as carryover resilience.

Lesson. Dimensional analysis would have caught this immediately. The product $B_i \times \text{tod}_t$ has units of (mmHg × weeks), which is not commensurate with the BR component (unitless symptom-score change). The resulting signal is orders of magnitude larger than the noise floor. Always check that the moderation term has the correct units and magnitude relative to the outcome variable.

3.4 Attempt 4: centered, binary, calibrated (correct)

The correct formulation standardizes the biomarker, uses a binary drug indicator, and scales by σ_{BR} to match the conditional expectation from the MVN model (Equation 3):

$$\text{BR}_{it}^{\text{mod}} = \text{BR}_{it}^{\text{base}} + c_{bm} \cdot \frac{B_i - \mu_B}{\sigma_B} \cdot \mathbf{1}[\text{on drug at } t] \cdot \sigma_{\text{BR}} \tag{11}$$

```
bm_z <- (dat$bm - bm_mean) / bm_sd
for (tp in 1:nP) {
  if (d[tp]$onDrug) {
    br_col <- paste(tnames[tp], "br", sep = ".")
    dat[, (br_col) := get(br_col) + c.bm * bm_z * br_sd]
  }
}
```

Why it works.

1. **Centering.** The standardized biomarker B_i^* has mean zero. The population-average moderation is zero; only deviations from the mean biomarker produce differential BR. This is the correct behavior: the intent is to model an *interaction* (differential response by biomarker level), not a main effect.

2. **Binary indicator.** The on-drug indicator $\mathbf{1}[\text{on drug}]$ is 1 during treatment and 0 during off-drug periods. Unlike tod (which grows with cumulative exposure) or $\phi(t)$ (which decays during washout), this indicator produces a clean, constant signal during all on-drug timepoints and zero signal during all off-drug timepoints. Carryover in the DGP affects the BR *means* but not the moderation term.

3. **Scaling.** The factor σ_{BR} ensures that c_{bm} has the same interpretation in both architectures: a one-SD shift in the biomarker produces a $c_{bm} \times \sigma_{\text{BR}}$ shift in BR. For $c_{bm} = 0.45$ and $\sigma_{\text{BR}} = 8$, this is 3.6 BR units, matching the Architecture B calibration target from Equation 5.

4 Empirical verification

4.1 Calibration

Drawing 500 participants under both architectures with identical parameters ($c_{bm} = 0.45$, $N = 500$, crossover design, no carryover), the regression of mean on-drug BR on the biomarker yields:

	Arch A	**Arch B**
BR shift per SD of biomarker	3.36	3.48
Residual SD	5.23	3.89

The shift magnitudes are comparable. Architecture A has slightly higher residual variance because the BM-BR correlation in the covariance matrix is absent (the noise components are independent), whereas Architecture B's MVN draw produces correlated noise that reduces the residual variance in the regression.

4.2 Power under carryover

Using the N-of-1 (Hybrid) design with $N = 70$, $c_{bm} = 0.45$, and 50 replicates per cell:

$t_{1/2}$ (weeks)	**Arch A**	**Arch B**	**Relative change (B)**
0.0	0.74	0.82	—
0.5	0.68	0.64	−22%
1.0	0.72	0.50	−39%

Architecture A power is stable (variation within Monte Carlo error at $N_{\text{reps}} = 50$). Architecture B shows progressive power loss with increasing carryover half-life, reaching a 39% relative decline at $t_{1/2} = 1.0$ week.

5 Discussion

5.1 Why scaling matters

The four attempts illustrate a general principle in simulation development: the *units and magnitude* of each component in the data-generating process must be verified against a known reference. Three of the four attempts produced code that executed without errors, generated plausible-looking data, and could be analyzed by the downstream statistical model. The errors were detectable only by checking whether the resulting power estimates were consistent with theoretical expectations.

- Attempt 1 was identifiable by checking the correlation between biomarker and BR during off-drug timepoints (expected: nonzero under Architecture A; observed: zero).
- Attempt 2 was identifiable by computing the signal-to-noise ratio at off-drug timepoints (expected: detectable; observed: negligible).
- Attempt 3 was identifiable by dimensional analysis (biomarker in mmHg, time in weeks, product in mmHg·weeks, outcome in symptom-score units) or by checking the signal-to-noise ratio (expected: ∼1; observed: ∼70).

5.2 The role of the conditional expectation

The correct formulation (Equation 11) was derived by asking: "What effect size does Architecture B produce for the same value of c_{bm}?" The answer comes directly from the conditional expectation of the MVN distribution (Equation 3). By matching this conditional expectation, the formulation ensures that $c_{bm} = 0.45$ means the same thing (in terms of detectable effect size) under both architectures. This is essential for the comparative power analysis that motivates the dual-architecture framework.

5.3 Implementation

The `dgp_architecture` parameter controls the choice between architectures in the `pmsimstats-ng` R package. It is threaded through three functions:

- `buildSigma()`: When `dgp_architecture = "mean_moderation"`, the BM-BR correlation block is omitted from the covariance matrix. The biomarker and BR components are drawn independently.
- `generateData()`: After the MVN draw, the additive moderation term (Equation 11) is applied to all on-drug BR columns.
- `generateSimulatedResults()`: The parameter is passed to the sigma cache and all `generateData()` calls within the simulation loop.

The default is `"mvn"` (Architecture B), preserving backward compatibility with the Hendrickson et al. (2020) publication code. All 67 existing unit tests pass under the default.

5.4 Known limitation: parameter validation

The `validateParameterGrid()` function, called at the start of `generateSimulatedResults()`, constructs sigma matrices using `buildSigma()` to check for positive definiteness before the simulation begins. This check currently uses the Architecture B covariance structure regardless of the `dgp_architecture` setting. Under Architecture A with $c_{bm} > 0.45$, the function will report spurious PD failures and print a warning, even though the actual simulation runs without issue because the BM-BR correlation block is omitted from the sigma matrix used for data generation.

This warning can be safely ignored when running Architecture A. A future revision should pass `dgp_architecture` to `validateParameterGrid()` so that the PD check uses the correct sigma structure.

6 Positive definiteness and the c_{bm} ceiling

6.1 The constraint under Architecture B

Under Architecture B, c_{bm} enters the off-diagonal entries of the covariance matrix as a correlation between the biomarker and each BR timepoint column. Combined with the within-factor autocorrelation ($\rho = 0.8$ in the publication), cross-factor correlations ($c_{\text{cf1t}} = 0.2$, $c_{\text{cfct}} = 0.1$), and the full set of timepoint columns, the matrix becomes non-positive-definite when c_{bm} exceeds approximately 0.45. The Hendrickson publication used $c_{bm} \in \{0, 0.3, 0.6\}$, but the $c_{bm} = 0.6$ condition fails PD checks for 7 of 9 trial paths at zero carryover and for all 9 paths at higher carryover values. The revised analysis in `04-revised-power-analysis.tex` was forced to reduce the grid to $c_{bm} \leq 0.45$ and the autocorrelation to 0.7.

6.2 Architecture A removes the constraint

Under Architecture A, the BM-BR correlation block is absent from the covariance matrix. The parameter c_{bm} enters only in the post-draw additive moderation step and does not affect the eigenvalue structure of Σ. Testing c_{bm} from 0 to 0.95 in increments of 0.05 across all four trial designs and three carryover half-lives:

	Arch B (MVN)	**Arch A (mean mod)**
First PD failure	$c_{bm} = 0.50$	None through 0.95
Feasible ceiling	∼0.45	Unconstrained
Paths failing at 0.60	7/9	0/9
Paths failing at 0.95	9/9	0/9

This is a direct consequence of removing the BM-BR entries from Σ: the matrix dimension and sparsity pattern are the same, but the off-diagonal entries that drove the smallest eigenvalues negative are gone.

6.3 Hendrickson publication parameters under Architecture A

With the PD constraint removed, it is possible to evaluate the original Hendrickson publication parameter grid: $c_{bm} \in \{0, 0.3, 0.6\}$, autocorrelation $\rho = 0.8$, carryover $t_{1/2} \in \{0, 0.1, 0.2\}$ weeks, $N \in \{35, 70\}$. Results at 50 replicates per cell, no censoring, analysis model without carryover term:

Design	N	$t_{1/2} = 0$	$t_{1/2} = 0.1$	$t_{1/2} = 0.2$
Power at $c_{bm} = 0.6$				
N-of-1	70	0.98	0.96	0.96
OL+BDC	70	0.92	0.92	0.92
CO	70	0.66	0.62	0.70
OL	70	0.14	0.16	0.20
N-of-1	35	0.68	0.78	0.70
OL+BDC	35	0.58	0.66	0.68
CO	35	0.40	0.28	0.26
Power at $c_{bm} = 0.3$				
N-of-1	70	0.48	0.38	0.50
OL+BDC	70	0.36	0.36	0.34
CO	70	0.26	0.20	0.28
Type I error ($c_{bm} = 0$)				
All designs	35–70	0.00–0.08 (nominal)		

Key observations:

1. **Carryover resilience is confirmed.** Power is stable across all three carryover half-lives for every design. The variation (e.g., N-of-1 at $N = 70$: 0.98, 0.96, 0.96) is within Monte Carlo error at 50 replicates.

2. **Design ranking is preserved.** N-of-1 > OL+BDC > CO > OL, consistent with the original Hendrickson findings and the revised Architecture B analysis.

3. $c_{bm} = 0.6$ **is now evaluable.** The N-of-1 design at $N = 70$ achieves 0.96–0.98 power, confirming that the original publication's target effect size is well-powered under this design. This result was previously unobtainable because the parameter failed PD validation under Architecture B.

4. **OL design has low power** even at $c_{bm} = 0.6$, because the all-on-drug design has no off-drug contrast for the `bm:Dbc` interaction term.

5. **Type I error is controlled** at 0–8% across all conditions, consistent with the nominal 5% level given Monte Carlo variability at 50 replicates.

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